



Operations Research

Prof. Dr. Martin Bichler,
Halil Bayrak, Janik Bürgermeister, Teodora Dobos,
Jan Höhener, Abheek Ghosh

Exercise Sheet 7

Demo Exercises

Exercise D7.1. *Cutting*

A carpentry shop has $20m$ and $17m$ boards at its disposal. It receives an order from a customer to produce 30 boards of $6m$ length (type A), 15 boards of $7m$ length (type B) and 40 boards of $3m$ length (type C). The use of a $20m$ board costs 20 euros, and the use of a $17m$ board costs 17 euros. The joinery wants to fulfill the order in such a way that the costs for it are minimal. It would like to solve this problem using column generation.

- For the CG algorithm, an optimal cutting pattern must be generated in each iteration with respect to a given utility vector $u = (u_A, u_B, u_C) \in \mathbb{R}^3$ and a given length $l \in \{17, 20\}$ of the initial board. Let u_T denote the utility of a board of type $T \in \{A, B, C\}$. Formulate an IP that solves the problem “find an optimal cut with respect to u and l ”.
- We choose as (bad) initial basis the 3 cutting patterns where exactly 1 board of one type is cut from a board of length $17m$. Run one iteration of the CG algorithm with this basis.

Exercise D7.2. *Vacation*

After an exhausting semester, the student Georgina plans to go on vacation with all her friends. In total, m people are going on vacation, with the i -th person wanting to bring $w_i \in \mathbb{N}$ kilograms of luggage. They plan to undertake the trip with several rental cars, where each person wants to travel in the car that transports their luggage, and the total weight of the luggage in the cars must not exceed $W \in \mathbb{N}$ kilograms each. Unfortunately, some of the friends have quarrels with each other, so they don't want to travel in the same car. Let $S \subseteq \{\{i, j\} \subseteq \{1, \dots, m\}, i \neq j\}$ denote the set of all quarrelsome pairs. Thus, $\{i, j\} \in S$ if and only if i does not want to sit in a car with j .

Of course, Georgina wants to rent as few cars as possible. She wants to approximate the optimal solution using Column Generation. To do this, she first formulates the following IP:

$$\begin{aligned} \min, & \sum_{i=1}^n c_i x_i \\ \text{s.t.} & Ax = b \\ & x \in \{0, 1\}^n. \end{aligned}$$

The matrix A has m rows and n columns, with each column describing a permissible "filling pattern" for a car. Furthermore, $c = (1, \dots, 1)^T \in \mathbb{R}^n$ and $b = (1, \dots, 1)^T \in \mathbb{R}^m$. The column $a \in \{0, 1\}^m$ describes a car in which exactly those persons j with $a_j = 1$ are sitting. $x_i = 1$ means that the filling pattern in the i -th column is used. The filling pattern must comply with the rules stated above. The allowable total weight W of the car must not be exceeded, and for no pair $\{i, j\} \in S$ should $a_i = 1$ and $a_j = 1$ hold. Furthermore, at least one person must travel in each used car. We call such a filling pattern *valid*.

- a) Describe the condition " $a = (a_1, \dots, a_m)^T \in \{0, 1\}^m$ is valid" through linear (in)equality constraints.
- b) In this subtask, we assume that $w_i = 10$ for all $i \in \{1, \dots, m\}$, $W = 40$, and $m \in \mathbb{N}$ is arbitrary. Specify the number n of columns of A for the two sets $S_1 = \emptyset$ and $S_2 = \{\{i, j\}, \dots, i \neq j\}$ of quarrelsome friends.
- c) Under what condition on the number $|S|$ of quarrelsome pairs is it particularly advisable to use the Column Generation Algorithm? Use the results from subtask (b) to justify your answer.
- d) For this subtask, we assume that $m = 4$ people are going on vacation, $w_i = 10$, $W = 40$, and $S = \{\{1, 2\}, \{2, 3\}\}$. Perform one iteration of the CG algorithm, starting with the basis in which each person travels alone in a car. Specify in particular the pricing problem and - with justification - the entering and exiting basic variables. Intermediate calculations need not be provided.